Splunk Brute Force Detection Lab

Authentication Log Analysis and Threat Detection

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Tool: Splunk Enterprise (local instance)

Dataset: Custom-built authentication log (auth_logs.csv)

Classification: Portfolio Project

1. EXECUTIVE SUMMARY

This project demonstrates brute force attack detection using Splunk and a custom authentication log dataset. The goal was to build a realistic investigation workflow from scratch: designing the dataset, loading it into Splunk, writing SPL queries to surface suspicious behavior, and interpreting the results the way a SOC analyst would.

A simulated authentication log was built with 15 events covering normal employee logins alongside a brute force attack scenario. The attacker IP (192.168.1.222) made seven failed login attempts against the account bdennis23 between 22:35 and 23:12, followed by a successful login. All SPL queries written during the investigation correctly identified the attacker based on behavioral patterns in the log data alone.

Item	Detail
Tool	Splunk Enterprise (free trial, local)
Dataset	Custom CSV - 15 authentication events
Attack Type	Brute force login attempt
Attacker IP	192.168.1.222
Target Account	bdennis23
Failed Attempts	7 (22:35 to 23:04)
Outcome	Attacker gained access at 23:12
SPL Commands Used	stats, sort, where, table

2. ENVIRONMENT SETUP

2.1 Tool

Splunk Enterprise was installed locally on a Windows desktop using the free 60-day trial. No external data sources or cloud services were used. The Splunk web interface was accessed at localhost:8000 throughout the project.

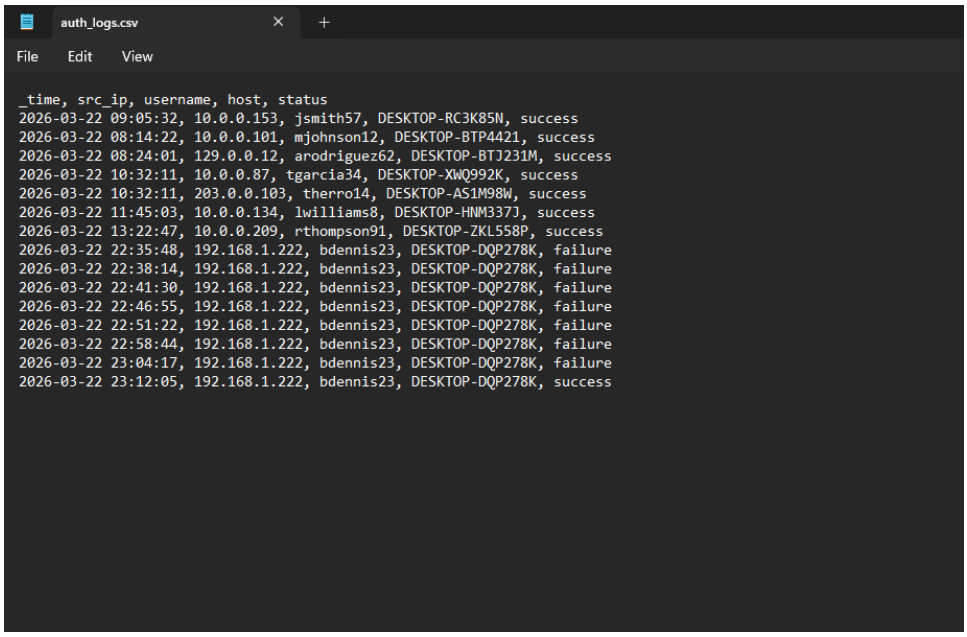
2.2 Dataset Design

Rather than using a large pre-built dataset, a small custom authentication log was designed from scratch. This kept the setup simple while ensuring the data contained exactly the patterns needed to demonstrate brute force detection.

The dataset was built as a CSV file with five fields:

- _time: timestamp of the login attempt
- src_ip: IP address the request came from
- username: account being targeted
- host: machine receiving the login attempt
- status: result of the attempt (success or failure)

The dataset contains 15 events: 7 normal successful logins from legitimate employees during business hours, 7 failed attempts from 192.168.1.222 against bdennis23 between 22:35 and 23:12, and 1 successful login from that IP after the failures.



auth_logs.csv

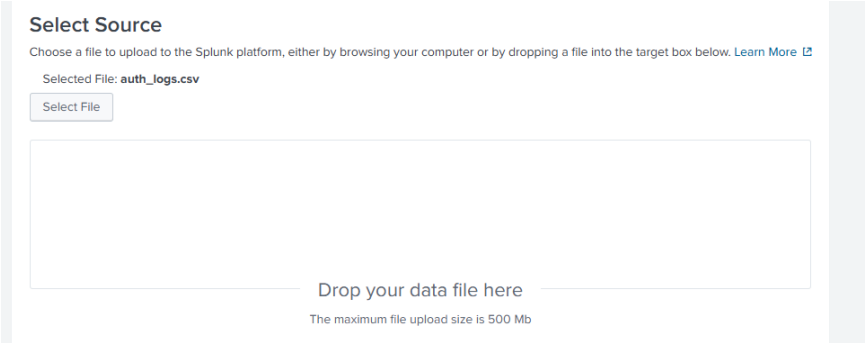
File Edit View

```
_time, src_ip, username, host, status
2026-03-22 09:05:32, 10.0.0.153, jsmith57, DESKTOP-RC3K85N, success
2026-03-22 08:14:22, 10.0.0.101, mjohnson12, DESKTOP-BTP4421, success
2026-03-22 08:24:01, 129.0.0.12, arodriguez62, DESKTOP-BTJ231M, success
2026-03-22 10:32:11, 10.0.0.87, tgarcia34, DESKTOP-XWQ992K, success
2026-03-22 10:32:11, 203.0.0.103, therro14, DESKTOP-AS1M98W, success
2026-03-22 11:45:03, 10.0.0.134, lwilliams8, DESKTOP-HNM337J, success
2026-03-22 13:22:47, 10.0.0.209, rthompson91, DESKTOP-ZKL558P, success
2026-03-22 22:35:48, 192.168.1.222, bdennis23, DESKTOP-DQP278K, failure
2026-03-22 22:38:14, 192.168.1.222, bdennis23, DESKTOP-DQP278K, failure
2026-03-22 22:41:30, 192.168.1.222, bdennis23, DESKTOP-DQP278K, failure
2026-03-22 22:46:55, 192.168.1.222, bdennis23, DESKTOP-DQP278K, failure
2026-03-22 22:51:22, 192.168.1.222, bdennis23, DESKTOP-DQP278K, failure
2026-03-22 22:58:44, 192.168.1.222, bdennis23, DESKTOP-DQP278K, failure
2026-03-22 23:04:17, 192.168.1.222, bdennis23, DESKTOP-DQP278K, failure
2026-03-22 23:12:05, 192.168.1.222, bdennis23, DESKTOP-DQP278K, success
```

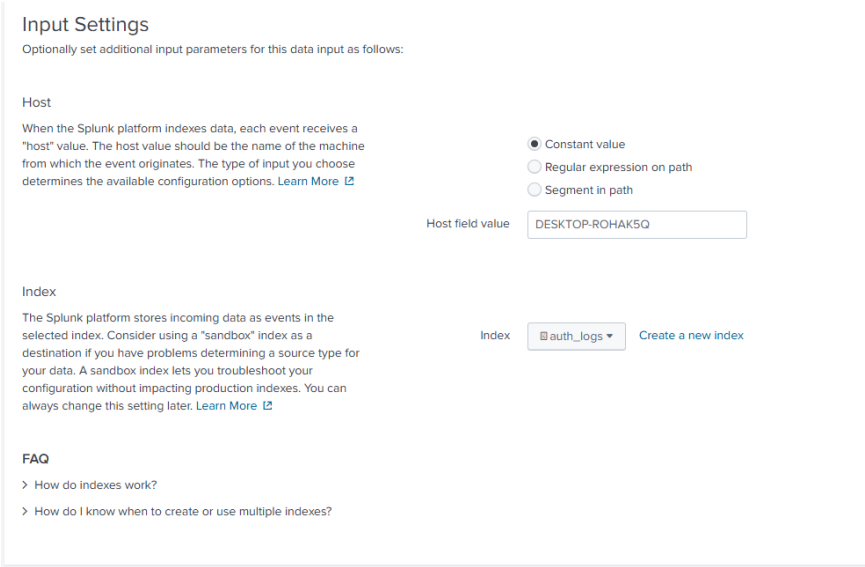
The auth_logs.csv file showing all 15 authentication events

2.3 Importing the Dataset

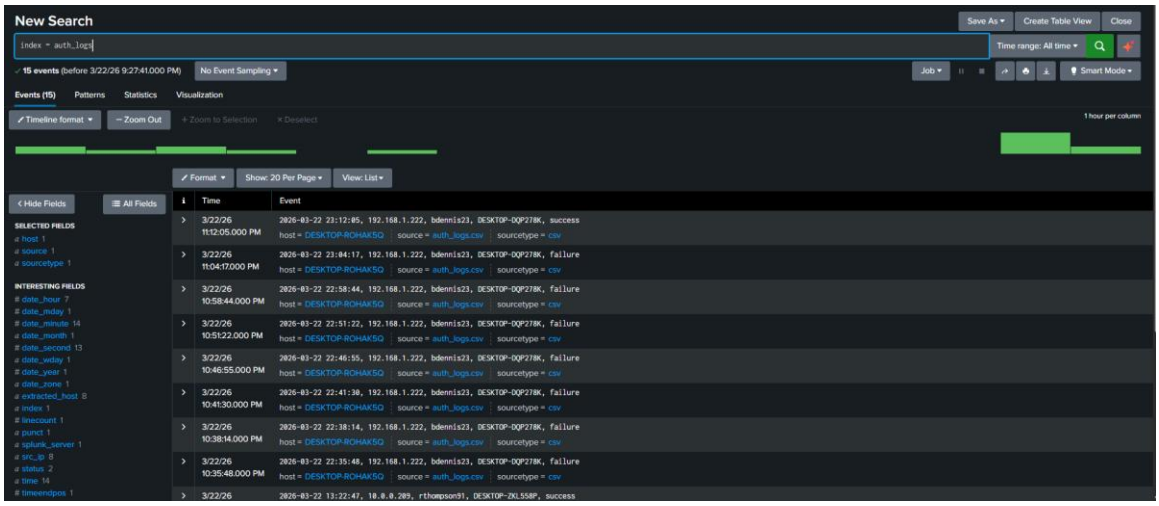
The CSV was imported through Settings > Add Data > Upload in Splunk. A dedicated index called auth_logs was created to keep the data separate from internal Splunk logs.



Splunk file upload screen with auth_logs.csv selected



Input settings showing the auth_logs index configured



All 15 events confirmed loaded into the auth_logs index (All Time)

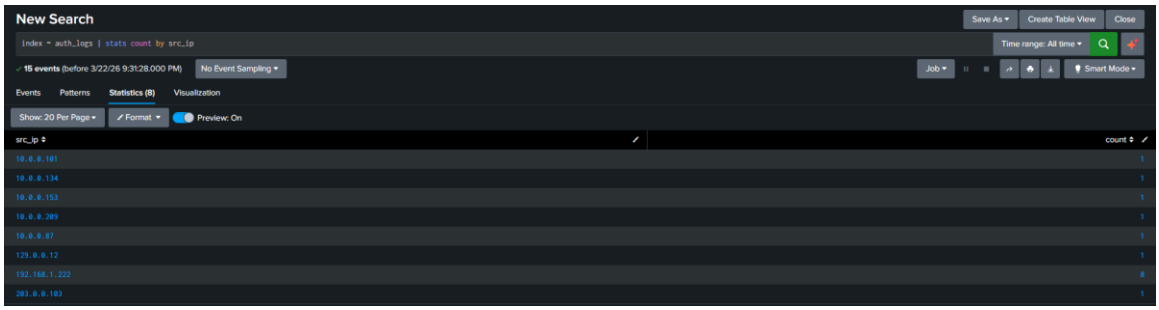
3. INVESTIGATION

The investigation follows the same general approach a SOC analyst takes: start broad to understand what is in the data, identify anomalies, narrow down to the suspicious activity, and reconstruct the timeline.

Query 1: Activity Overview by Source IP

```
index = auth_logs | stats count by src_ip
```

This query counts total events for each source IP. The goal is to get a high-level picture of who is generating traffic before any filtering. In a real environment this is typically the starting point when triaging a login-related alert.



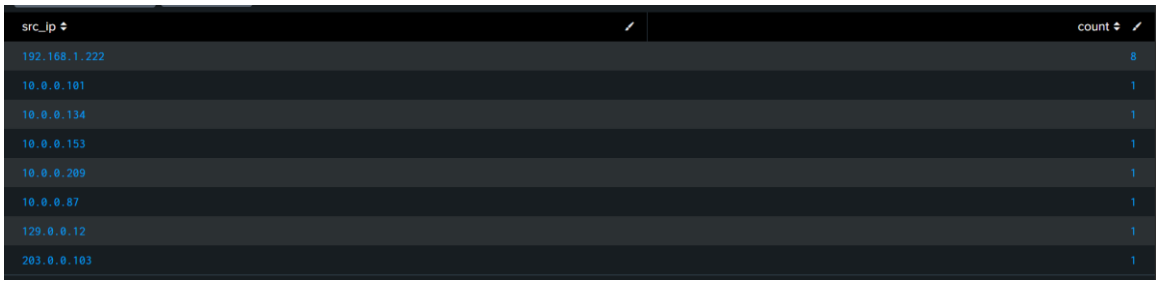
Event count per source IP. All legitimate users have 1 event. 192.168.1.222 has 8.

Every legitimate employee has exactly one event, which is expected. The IP 192.168.1.222 has 8 events, which stands out immediately and warrants investigation.

Query 2: Sorting by Event Count

```
index = auth_logs | stats count by src_ip | sort -count
```

Adding sort -count reorders results from highest to lowest, putting the most active IPs at the top. In a real log with thousands of source IPs this step is what makes the outlier visible without scrolling through everything.



src_ip	count
192.168.1.222	8
10.0.0.101	1
10.0.0.134	1
10.0.0.153	1
10.0.0.209	1
10.0.0.87	1
129.0.0.12	1
203.0.0.103	1

192.168.1.222 rises to the top with 8 events. All other IPs have 1.

Query 3: Breaking Down Activity by Status

```
index = auth_logs | stats count by src_ip, status
```

Grouping by both src_ip and status shows whether activity from a suspicious IP is mostly failures, mostly successes, or a mix. A high failure count followed by a success is the signature pattern of a brute force attack.

src_ip	status	count
10.0.0.101	success	1
10.0.0.134	success	1
10.0.0.153	success	1
10.0.0.209	success	1
10.0.0.87	success	1
129.0.0.12	success	1
192.168.1.222	failure	7
192.168.1.222	success	1
203.0.0.103	success	1

Every legitimate IP shows a single success. 192.168.1.222 shows 7 failures and 1 success.

This confirms the brute force pattern. The attacker failed seven times before succeeding, which is consistent with an automated credential attack cycling through a password list until one works.

Query 4: Automated Brute Force Detection

```
index = auth_logs status = failure | stats count by src_ip | where count
> 3 | sort -count | table src_ip, count
```

This query automates the detection by applying a threshold. Only IPs with more than 3 failed login attempts are returned. The threshold can be adjusted per environment. In a real SOC this query would be saved as an alert that fires automatically when the condition is met.

```
index = auth_logs status = failure | stats count by src_ip | where count > 3 | sort -count
```

✓ **7 events** (before 3/22/26 9:51:44.000 PM) No Event Sampling ▾

Brute force detection query - only IPs with more than 3 failed attempts are returned

src_ip	count
192.168.1.222	7

192.168.1.222 is the only IP flagged, with 7 failed attempts

The query returns exactly one result. No legitimate users are flagged. This is a clean detection with no false positives in this dataset.

Query 5: Reconstructing the Attack Timeline

```
index = auth_logs src_ip = 192.168.1.222 | sort _time | table _time,
src_ip, username, host, status
```

With the attacker IP identified, this query pulls every event from that IP in chronological order. This is the timeline reconstruction step, showing when the attack started, how long it ran, and when access was gained.

_time	src_ip	username	host	status
2026-03-22 22:35:48	192.168.1.222	bdennis23	DESKTOP-ROHAK5Q	failure
2026-03-22 22:38:14	192.168.1.222	bdennis23	DESKTOP-ROHAK5Q	failure
2026-03-22 22:41:30	192.168.1.222	bdennis23	DESKTOP-ROHAK5Q	failure
2026-03-22 22:46:55	192.168.1.222	bdennis23	DESKTOP-ROHAK5Q	failure
2026-03-22 22:51:22	192.168.1.222	bdennis23	DESKTOP-ROHAK5Q	failure
2026-03-22 22:58:44	192.168.1.222	bdennis23	DESKTOP-ROHAK5Q	failure
2026-03-22 23:04:17	192.168.1.222	bdennis23	DESKTOP-ROHAK5Q	failure
2026-03-22 23:12:05	192.168.1.222	bdennis23	DESKTOP-ROHAK5Q	success

Full attack timeline: 7 failures from 22:35 to 23:04, successful login at 23:12

4. FINDINGS

Based on the investigation, the following was determined:

Brute Force Attack Against Account bdennis23

The IP address 192.168.1.222 conducted a brute force attack against bdennis23 on March 22, 2026. The attack began at 22:35 and the attacker gained access at 23:12 after 7 failed attempts over 37 minutes.

Attacker IP	192.168.1.222
Target Account	bdennis23
Target Host	DESKTOP-DQP278K
Attack Start	2026-03-22 22:35:48
Account Compromised	2026-03-22 23:12:05
Duration	37 minutes
Failed Attempts	7
Attack Window	Late night, outside normal business hours

Two additional behavioral indicators support this finding:

- Timing: all attack activity occurred between 22:35 and 23:12, well outside the normal business hours window where legitimate logins were observed (08:00 to 13:30).
- Pattern: every other source IP generated exactly one successful login. 192.168.1.222 was the only IP with repeated failures followed by a success.

5. RECOMMENDATIONS

Account Lockout Policy

Implement account lockout after a set number of failed attempts (typically 5 to 10). This would have stopped the attack before the attacker succeeded. Active Directory and most identity providers support this natively.

Multi-Factor Authentication

Even if a password is guessed correctly, MFA prevents access without a second factor. This is the most effective control against brute force attacks and would have blocked the successful login at 23:12 regardless of whether the password was cracked.

Automated Alerting in Splunk

The detection query from Query 4 can be saved as a Splunk scheduled alert that fires when any IP crosses the failure threshold. This removes manual investigation from the equation and reduces time to detection significantly.

After-Hours Login Monitoring

The attack ran from 22:35 to 23:12. Alerts scoped to login failures outside business hours would have flagged this before the account was compromised.

6. CONCLUSION

This project covered the full detection workflow for a brute force attack using Splunk: designing a realistic dataset, loading it into a local instance, writing SPL queries to surface behavioral anomalies, and interpreting the results to reconstruct what happened.

The four SPL commands used throughout the investigation, stats, sort, where, and table, are the foundation of most real-world log analysis work. Understanding how to chain them together to go from raw data to a specific finding is the core skill this project was built around.

The attacker IP was identified without any prior knowledge of who the attacker was. The detection came purely from the data showing an abnormal pattern. That is what makes SPL useful in a real SOC environment, where you are often looking for something without knowing in advance what you are looking for.

7. REFERENCES

Splunk Enterprise Free Trial: https://www.splunk.com/en_us/download/splunk-enterprise.html

Splunk SPL Documentation: https://docs.splunk.com/Documentation/Splunk/latest/SearchReference

OWASP Brute Force Attack: https://owasp.org/www-community/attacks/Brute_force_attack

MITRE ATT&CK T1110 - Brute Force: https://attack.mitre.org/techniques/T1110/